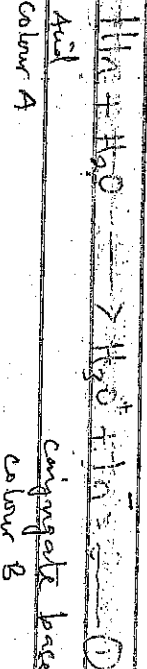


Acid-Base Titration

An acid-base titration is a laboratory procedure used in qualitative chemical analysis to determine the concentration of either a base or an acid. It is often carried out using a visual indicator to determine the end-point. An acid of known concentration is added from a burette to a known volume (base) of unknown concentration in a conical flask. The end point occurs when stoichiometric amount of the acid has been added to the base at this point complete neutralization is achieved. The solution consists of salt and water only. From the values of the molarity of the acid. The volume of acid used and the volume of base used. The molarity of the base can be determined.

Indicators (Acid-Base)

An acid-base indicator is a laboratory indicator that is a substance which has one colour in an acid & another colour in a base. They are usually weak acids or bases. Acid-base indicators are also known as pH indicators. Consider an indicator which is a weak acid with general formula HIn . When they dissolved in water, an equilibrium is established between the weak acid & its conjugate base.



We can apply the equilibrium law to the above expression to generate for a weak acid HIn , the dissociation constant is given by.

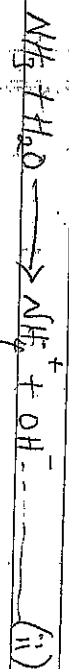
$$K_{In} = \frac{[H_3O^+][In^-]}{[HIn]} \quad (2)$$

where K_{In} is called the indicator dissociation constant. The colour of an indicator changes from A to B or vice versa at a point called the turning point. At this point the concentration of the acid HIn is equal to the concentration of the conjugate base In^- .

Can be prepared from a weak base & one of its salt. A solution of NH_3 is typically used for this purpose. In aqueous solution, ammonium chloride dissociates into:



NH_3 is a weak base, partially dissociates into



If acid is added, it is neutralised by hydroxide ions. As a result, equilibrium of equation (ii) shifts to the right. The shift maintains a constant concentration of hydroxide ions & thus a constant pH.

If a base is added, equilibrium of equation (ii) shifts to the left, thus maintaining the concentration of OH^- in constant. The presence of NH_4Cl in the buffer solution ensures that there is sufficient NH_4^+ to cope with additions of base. The Henderson equation for a buffer solution made up of a weak base & one of its salt is given by

$$\text{pH} = \text{pK}_a - \text{pK}_b + \log \left[\frac{\text{base}}{\text{acid}} \right]$$

(iii) Strong Base Buffer

A strong base can be used as a buffer solution with a pH value. The addition of a small amount of acid or base has negligible effect on the pH e.g. when 1 cm³ of 0.1 M of HCl is added to 100 cm³ of 0.1 mol/dm³ NaOH solution. The pH drops from 12 to 11.94. This drop of 0.04 can be checked using the equation for calculation of the pH of the base before and after the addition of the acid. For a strong base to determine the pH we use the equation:

(iv) Strong Acid Buffer

A strong acid such as HNO_3 can act as a buffer with low pH. Strong acids are fully dissociated in aqueous solution. Thus, the concentration of H_3O^+ ions is high. The addition

(14)

$$= 0.25 - 2$$

$$\log[\text{salt}] = -1.75$$

finding the ⁰negative

$$\text{salt conc.} = -\log[1.75]$$

$$= 0.0178 = 1.78 \times 10^{-2}$$

$$\text{No. of moles} = \frac{\text{mass}}{\text{molar mass}}$$

molar mass

$$\text{but molar mass of } \text{CH}_3\text{COOH} = 82 \text{ g mol}^{-1}$$

$$\text{If 1 mole of } \text{CH}_3\text{COOH} = 82 \text{ g/mol}$$

$$\text{then } 1.78 \times 10^{-2}$$

making X subject of the formula

$$X = 1.458 \approx 1.46 \text{ g}$$

Henderson Hasselbalch equation

for a buffer solution of a weak acid & its conjugate base

$$\text{pH} = \text{pK}_a + \log \frac{[\text{conjugate base}]}{[\text{acid}]}$$

for a buffer solution of a weak base & its conjugate acid

$$\text{pOH} = \text{pK}_a + \log \frac{[\text{conjugate base}]}{[\text{acid}]}$$

Example:

Calculate the pH of a solution containing 0.2 mol/dm^3 CH_3COOH & 0.3 mol/dm^3 CH_3COO^- (K_a of $\text{CH}_3\text{COOH} = 1.8 \times 10^{-5} \text{ mol/dm}^3$)

Answer:

for a buffer mixture of CH_3COOH & CH_3COO^-

$$\text{pH} = \text{pK}_a + \log \frac{[\text{conjugate base}]}{[\text{acid}]}$$

$$\text{conjugate base} \neq (\text{CH}_3\text{COO}^-) = 0.3 \text{ mol/dm}^3$$

$$\text{acid} (\text{CH}_3\text{COOH}) = 0.2 \text{ mol/dm}^3$$

$$\text{pH} = -\log(1.8 \times 10^{-5}) + \log \frac{[0.3]}{[0.2]}$$

$$= -(4.74) + 0.18$$

$$= 4.92 //$$

(16)